

Mathematics & Statistics
MATH 2Z03
Engineering Mathematics III
Fall 2023



SCIENCE

Instructor Information



Professor David Earn

Email: earn@mcmaster.ca

Office: Hamilton Hall 317

Office Hours:

(tentative) Fri 11:30am–12:20pm, or by appointment

Web site: <https://davidearn.mcmaster.ca/>



Professor Gail Wolkowicz

Email: wolkowic@mcmaster.ca

Office: Hamilton Hall 318

Office Hours:

(tentative) Wed 12:30pm–1:20pm or by appointment

Web site: <https://ms.mcmaster.ca/wolkowic/>

Course Information

All course announcements will be posted on [Avenue to Learn](#).

Course Dates: 09/05/2023 - 12/06/2023

Units: 3.00

Course Delivery Mode: In Person

Course Description: Ordinary differential equations, Laplace transforms, introduction to partial differential equations, applications. Three lectures, one tutorial, one lab (two hours) every other week; one term Prerequisite(s): MATH 1ZB3 or 1ZC3 Prerequisite(s) (EFFECTIVE 2020-2021): MATH 1ZB3 Antirequisite(s): ENGINEER 2Z03, MATH 2C03, 2M03, 2P04

Instructor-Specific Course Information

All **lectures** will be in [Togo Salmon Hall 120](#).

- Section **C01, Dr. Earn**: Mon,Wed @ 8:30am, Fri @ 10:30am.
- Section **C02, Dr. Wolkowicz**: Mon,Wed @ 11:30am, Fri @ 1:30pm.

Weekly **tutorials**, led by **Elkin Ramirez** <ramireze@mcmaster.ca>, will begin in the second week of classes:

- Section **T01**: Wed @ 9:30am in [TSH 120](#).
- Section **T02**: Fri @ 12.30pm in [LRW B1007](#).

Biweekly **labs** (5 during the term) on [childsmath](#) will be due on Tuesdays @ 11:59pm, beginning 26 September 2023. You are encouraged to try the labs on your own, but if you would like help, there will be a Math 2Z03 TA available to help in two computer labs ([BSB 244](#) and [KTH B121](#)) from 2:30pm to 8:30pm on Mondays and Tuesdays before each lab is due, i.e.,

- September: 25, 26
- October: 16, 17 and 30, 31
- November: 13, 14 and 27, 28

Important Links

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)
- [eReserves](#)

- [The latest version of this document on the Simple Syllabus web site.](#)

Course Learning Outcomes

- Students are expected to learn how to solve first-order and second-order, ordinary and partial differential equations by using separation of variables, the linear superposition principle, and Laplace transform.

Required Materials and Texts

Textbook Listing: <https://textbooks.mcmaster.ca>



Advanced Engineering Mathematics

ISBN: 978-1-284-25008-4

Authors: Dennis G. Zill

Publisher: Jones and Bartlett Learning

Publication Date: 2022

Edition: 7th

Package with solutions manual.

Class Format

In Person

Lectures will also be available for live-stream, and as recordings approximately 24 hours later. See the Echo360 section on the Avenue course page.

Course Evaluation

Assignments and labs will be due in alternate weeks, beginning the third week of classes. There will be two mid-term tests, one in late October and one in late November; dates and times will be announced on the Avenue course page. There will be a 2.5 hour final exam, scheduled by the registrar. Whichever of the following grading schemes yields a higher mark will be used to compute your final grade.

Grading Scheme 1		Grading Scheme 2	
Best 5 of 6 Online Assignments	10%	Best 5 of 6 Online Assignments	10%

Best 4 of 5 Computer Labs	10%	Best 4 of 5 Computer Labs	10%
2 mid-term tests	20% each	Best 1 of 2 mid-term tests	25%
Final examination	40%	Final examination	55%

Note: If you miss one mid-term test then your grade will automatically be calculated using Grading scheme 2. In addition, for either grading scheme, if a mid-term test mark is lower than the final exam mark, then it will be replaced by the final exam mark. However, at least one mid-term test is mandatory: if you miss *both* scheduled mid-term tests then you *must* e-mail your instructor immediately and you will be given an oral test worth 25% of your grade instead.

Course Evaluation Details

The following are **tentative** dates for the two mid-term tests. You must e-mail your instructor immediately if you have a conflict or are unable to write at these times. Note that if these test dates or times change, you are responsible for informing your instructor immediately if you are unable to write at the scheduled time.

- **Test #1:** Wed 25 October 2023 (**evening**, 6:30pm or later)
- **Test #2:** Wed 29 November 2023 (**evening**, 6:30pm or later)

Grading Scale

Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69

Grade	Equivalent Grade Point	Equivalent Percentages
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

Course Schedule

MATH 2Z03 (Fall 2023)

	M	D	Tentative Topic	Assessment	Due/Test date
Week 1					
Lecture 1	Sep	6	1.1 Definitions and terminology		
Lecture 2		8	1.2 Initial Value Problems		
Week 2					
Lecture 3		11	2.1 Direction Fields and Autonomous Equations		
Lecture 4		13	2.1 (cont'd)		
Lecture 5		15	2.2 Separable Equations		
Week 3					
Lecture 6		18	2.3 Linear Equations	Assignment 1	Tues, Sep 19
Lecture 7		20	2.3 (cont'd)		
Lecture 8		22	2.7 Linear Models (<i>inc. Exp. Growth & Decay, Newton's Cooling</i>)		

Note: The above schedule is obviously not yet complete. It will be updated during the term.

Late Assignments

All assignments will be available on [childsmath](https://childsmath.com) and must be completed on time.

Solutions will be discussed in tutorials immediately after the due date, so late assignments cannot be considered.

Turnitin.com

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Generative AI: Use Prohibited

Students are not permitted to use generative AI in this course. In alignment with [McMaster academic integrity policy](#), it “shall be an offence knowingly to ... submit academic work for assessment that was purchased or acquired from another source”. This includes work created by generative AI tools. Also state in the policy is the following, “Contract Cheating is the act of “outsourcing of student work to third parties” (Lancaster & Clarke, 2016, p. 639) with or without payment.” Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/), located at <https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/>

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity / Plagiarism

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

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Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they

access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online.**

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's [Academic Accommodation of Students with Disabilities](#) policy.

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware

that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.

Academic Support for Students in the Faculty of Science

Students in the Faculty of Science can seek academic support and advice from the Office of the Associate Dean of Science (Academic). Our team is committed to your success! When you need support, have a question, or need direction – please contact the Office of the Associate Dean (Academic). The knowledgeable, friendly team is here to support your academic success during your undergraduate studies with the Faculty of Science.

Our office is located in Burke Science Building (BSB) Room 136. We can be reached online at <https://science.mcmaster.ca/associatedean/current-students/academic-advising.html>. Students are also encouraged to familiarize themselves with policies related to [Academic Accommodations for Students with Disabilities](#) and [Academic Accommodations for Religious, Indigenous or Spiritual Observances](#).

Course Specific Relief for Missed Academic Work Information

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

INSTRUCTOR TO ADD COURSE SPECIFIC INFORMATION HERE

Checking Grades & Regrade Policy

Checking Your Grades

Your marks will be recorded on Avenue. It is your responsibility to check that all grades entered into Avenue are recorded properly. All grade concerns and discrepancies must be reported to the Instructional Assistant within a week of receiving the grade.

Regrade Policy

Office hours with the TAs (and or instructor/IA) will be arranged following assessments to address concerns students may have regarding their grade. Requests for re-evaluation of tests or assignments must be made in writing to the IA using the form provided in Avenue and justifying the reason for which the assignment should be remarked. This should be done within one week of discussing the marked test or assignment with the instructional team and BEFORE the final exam is written. If the IA considers the written justification to be insufficient (e.g. simply wanting a higher grade is insufficient), the assignment will not be re-graded. If the justification is considered sufficient, the entire assignment/test will be re-graded. Please be aware that this may result in an increase, decrease or no change to the original mark. There will be no alternative assignments that can be completed by students for the purpose of increasing their final grade.